Energy Conversion and Management 332 (2025) 119707

Contents lists available at ScienceDirect

Energy Conversion and Management

journal homepage: www.elsevier.com/locate/enconman

# Chance constrained optimal sizing of a hybrid PV/battery/hydrogen isolated microgrid: A life-cycle analysis

Ahmed R. El Shamy *, Prince Aduama, Ameena S. Al-Sumaiti *

*Smart OR Lab, Advanced Power and Energy Center, Department of Electrical Engineering, Khalifa University, Abu Dhabi, P.O. Box 127788, United Arab Emirates*

ARTICLE INFO

*Keywords:*
Batteries
Chance constrained optimization
Hydrogen storage
Isolated microgrid
Photovoltaics
Sizing

ABSTRACT

Isolated microgrids offer a great potential for capturing advancements in renewable energy sources (RES) and storage technologies. Prospects for such systems include industrial, commercial, and residential areas, where grid connection is not possible or not economical. Planning isolated microgrids must be accurate, such that it preserves system security and avoids resource oversizing. This paper introduces a microgrid, composed of multiple storage components and a solar PV system, with the goal to optimally size the PV plant, battery capacity, fuel cell rating, electrolyzer rating and hydrogen tank size to match the load demand and reduce the life cycle cost of the entire system. A PV plant is used as the main energy source, while battery and hydrogen storage systems provide the maneuvering needed for covering the periods of sun absence. To accommodate PV and load uncertainties due to weather conditions, a chance constrained (CC) optimal sizing mixed integer linear programming (MILP) problem is formulated. CC optimization provides a great window for stochasticity inclusion and sizing based on criticality level of application. To prove the fidelity of the results, the system is required to achieve at least 80 % success of 100 different scenarios generated for the modeled system. The results show the capacities of the various components needed to meet the load requirements of a microgrid with a PV source, hydrogen storage system and a battery energy storage system. The total life cycle cost of the system for a period of 25 years is found to be US$1.221 m.

## 1. Introduction

It is estimated that about a billion people live without electricity globally, with majority in rural areas [1]. Renewable energy sources (RES) can help greatly in reducing that number. So, to accommodate RES in modern power systems, microgrid systems have become more popular. These microgrid systems have the potential of addressing the problem of electricity access. However, RES have intermittent nature [2], and their performance is very difficult to take for granted, which is usually the case with traditional bulky power grid [3,4]. This led to huge efforts in improving energy storage technologies for the purpose of achieving a 100 % renewable grid [5,6]. The presence of an energy storage system along with photovoltaic (PV) systems is essential when the microgrid in operating in standalone mode [7]. Most common types of energy storage are battery energy storage systems (BESSs). Recently, hydrogen energy storage systems have gained great attention in both power grid and transportation applications. The advantages of hydrogen storage includes their use flexibility, and faster charging compared to BESSs [8,9]. The end goal of all ongoing development in energy storage systems and RES and imposed regulations is to cut carbon resources share in energy sectors to zero [10,11]. Standalone/isolated microgrids are small scale utilities that give a glimpse on the needed future grids. Expected challenges for achieving 100 % renewables can be mitigated by studying isolated microgrids that operate on smaller scales. A main challenge is the sizing of components for the standalone systems [12]. On one hand, the system is required to be largely secure, as any initial planning miss-consideration can lead to energy deficit. On the other hand, oversizing of RES and storage systems may lead to an unnecessary investment, increased project footprint and energy waste [13,14]. For these reasons, many works have been done in the sizing and selection of standalone microgrids equipment. The issue can be formulated as an optimization problem with several objectives, such as extending system lifetime, increasing system reliability and minimizing investment cost [13,15]. Others such as [16] focuses on minimizing life cycle costs as well as the environmental impacts, meeting the energy and heating needs [17,18]. A review of some of the works found in the literature on sizing hybrid microgrid components is presented below.

For standalone microgrids, some research consider only RES and

* Corresponding authors.
*E-mail addresses:* 100062060@ku.ac.ae (A.R. El Shamy), ameena.alsumaiti@ku.ac.ae (A.S. Al-Sumaiti).

https://doi.org/10.1016/j.enconman.2025.119707
Received 4 November 2024; Received in revised form 17 January 2025; Accepted 4 March 2025
Available online 15 March 2025
0196-8904/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

storage systems [19], while others include traditional carbon-based sources, such as diesel generators [20]. A necessary requirement for problems involving RES is inclusion of the inherent uncertainty [21]. Some used historical data to generate probabilistic distributions to be used for obtaining the optimization samples [22]. Others depend on running scenario-based optimizations for modelling system uncertainties [23,24]. Robust optimizations are also used for this purpose, where a range of the most probable random quantities is found, and a worst case scenario approach is utilized for the sizing process [25]. This approach can lead to oversizing as worst case scenarios might have a very low probability, but optimizers are forced to cope with them [26]. Multi-objective frameworks are adopted in research such as [19]. In [27], the multi-objective formulation is utilized for sizing RES and energy storage in an isolated microgrid. The objectives used were minimizing capital investment cost and load shedding risk, considering RES and demand uncertainties. A robust shortfall risk is implemented with a Wasserstein ambiguity set. In [28], the multi-objective framework is implemented in designing hybrid solar power systems that combine concentrated solar power (CSP) with PV technologies. The goal of this approach was to determine the best trade-offs between cost and environmental impact. An adaptive robust optimization approach is employed in [29] to plan New York State's energy transition under various uncertainties such volatility in wind and solar power generation and market availability. A robust optimization to account for uncertainties in resources, costs, and discount rates is presented in [30]. Through this optimization, the authors developed an integrated system using hydrogen storage and hybrid renewable energy sources to supply both electricity and hydrogen. A deterministic approach was used in [31] to size a standalone grid with PV, BESS and a biogas power plant. Sizing was based on historical irradiance data, and uncertainty effect in the sizing process are considered by creating a histogram from the results of the daily sizing problems. Different combinations of hybrid PV/wind/battery/on-grid/off-grid systems are investigated in [19]. The authors perform sensitivity analysis involving electricity price, discount rates and demand to investigate the different sizing probabilities. A reliability-oriented methodology for sizing of a PV/wind and fuel cells is introduced in [32] with equivalent loss factor representing the reliability index constraint. A multi-objective planning of distribution system with RES is performed in [32]. Stochastic scenarios for uncertain variables were found and the optimal sizes were obtained by aggregating the probability-weighted results of each scenario. Objectives of this study included minimizing system investment, emissions, power losses and voltage deviations. The authors in [33] introduced a hierarchical energy management system for off-grid microgrids to improve the real-time balancing of microgrid generation and load, incorporating multiple distributed energy resources. The hierarchy consisted of mixed-integer linear programming (MILP) optimization for day-ahead scheduling of energy resources and real time stochastic model predictive control handling forecast errors and uncertainties. A two-stage day-ahead robust optimization and real-time rolling optimization is presented in [34]. In this design, a hydrogen energy storage system and carbon capture and utilization system is utilized to enhance renewable energy use and reduce carbon emissions. A hybrid optimization approach combining particle swarm optimization and genetic algorithm is utilized to optimize system capacities of PV, wind, battery, electrolyzer, hydrogen tank, and fuel cell in [15]. The model focused on minimizing the weighted average cost of energy while achieving renewable energy fraction within the system.

Chance constrained (CC) optimization approaches have been gaining much interest recently. This is because of their inherent capability of being adapted to different sizing criticality levels [35]. Solving CC problems analytically is usually very challenging. Scenario-based CC proposes a great potential for solving these problems, although the formulation becomes exhaustive [36]. Robust sets might be used for decreasing the number of scenarios involved, especially in problems with high random variables dimension [37]. CC optimization was used in [38] to design the capacities of an islanded microgrid with PV and wind energy systems and BESS. A robust set was introduced due to involving three uncertain variables. In [39], a two-stage CC framework is implemented for sizing and operation dispatching of PV-based power system. In [40], genetic algorithm is used with Monte-Carlo simulation for running CC optimization scenarios. The target in [40] is sizing distributed generation units in a distribution network, imposing security constraints involving current limits. A distributionally robust formulation is performed in [41] to guarantee robust results of the planning and expansion of distribution networks involving RES.

This paper proposes a sizing problem for a hybrid standalone microgrid. The proposed microgrid consists of a PV plant, BESS, and a hydrogen storage system. It is required to find the size of PV plant, BESS, fuel cell generator, electrolyzer and hydrogen tank that would minimize the total system cost over the system's lifetime. These costs include the total capital costs of the system components, the cost of replacement of system components, and the operational and maintenance costs. In addition, security constraints are achieved intrinsically by preserving the limits of stored energy, that may otherwise be used in extreme conditions. A major issue for sizing PV-based systems is their stochastic behaviour, along with load variability. For this purpose, a CC optimization approach is used. CC optimization provides a good approach for excluding extreme scenarios in problems involving uncertainties. That is very useful for the proposed model where there is always some amount of energy stored to provide some security margin. The problem is formulated as scenario-based, where random number of scenarios are generated to achieve a required level of criticality to ensure safe design of equipment. Due to the complexity of CC formulations and involving large amounts of decision variables, the sizing is based on 24-hour operational scenarios for both solar irradiance and load historical data. To account for uncertainties, copula is used in this work. Copula enables measuring and modelling of the associations between variables while allowing for flexibility in modelling different types of dependencies. The problem constraints are designed to avoid long-term BESS degradation by imposing limits on its rate of use. Fuel cells efficiency becomes very low at lighter loading conditions. Therefore, limits are imposed on usage of fuel cells to guarantee their efficiency and obtaining realistic sizing.

The main contributions of this paper can be summarized as follows:

1. Developing a microgrid with multiple storage systems that offer flexibility to the overall system with consideration of its total system costs over its lifetime.
2. Implementing a chance constrained optimization approach for a microgrid with a with PV, BESS and hydrogen (including fuel cell, electrolyzer and hydrogen tank) systems, and the constraints formulation to guarantee minimum cycling of BESS for preserving its lifetime. Moreover, the efficiency of both storage systems is implicitly considered by limiting the rate of charging/discharging of battery and rate of stored hydrogen energy use.

Furthermore, Table 1 compares this work with other works within the area of hybrid RES storage models.

The work is organized as follows. Section 2 introduces the description and modelling of all subsystems used in the microgrid. The CC methodology and problem formulation is discussed in section 3. Then, the results and related discussions are presented in section 4. Finally, the conclusion is offered in section 5.

## 2. System description and modeling

The proposed microgrid consists of a PV plant, BESS, a hydrogen storage system, and an AC load. The different nature of both storage systems (battery and hydrogen storage system) provides more flexibility for microgrid operation. Schematic diagram of the proposed microgrid system is shown in Fig. 1. The PV plant acts as the main energy source.

**Table 1**
Comparison of proposed approach with other models.

| Reference | RES | Optimization approach | Uncertainty characterization | Economic analysis | Uncertainty Scenario generation | Results corroboration with other solvers |
|---|---|---|---|---|---|---|
| [15] | PV/BESS/ Hydrogen system | Particle swarm optimization (PSO) and genetic algorithm (GA) | No | Yes | Six cases (author defined) | No |
| [42] | PV/ Hydrogen system | hybrid artificial bee colony (ABC)-PSO | No | Yes | One case (author defined) | Yes |
| [43] | Wind/PV/ Hydrogen system | PSO | No | Yes | One case (author defined) | No |
| [44] | PV/BESS/ Hydrogen system | PSO, Whale Optimization Algorithm (WOA), ant colony optimization (ACO), GA | No | Yes | Two cases (author defined) | Yes |
| [45] | PV/BESS/ Hydrogen system | Robust Optimization | Yes | Yes | Robust optimization (uncertainty sets) | No |
| [46] | PV/BESS/ Hydrogen system | Robust Optimization | Yes | Yes | Sparse PCE | No |
| [47] | | Marine Predators Algorithm (MPA) and Seagull Optimization Algorithm (SOA) | Partial (limited to three different load %) | Yes | No | Yes |
| [48] | PV/BESS/ Hydrogen system | Information Gap Decision Theory (IGDT) | Partial (limited to load) | Yes | Information Gap Decision Theory (IGDT) | No |
| This work | PV/BESS/ Hydrogen system | CC optimization | Yes | Yes | 100 scenarios using copulas | Yes |

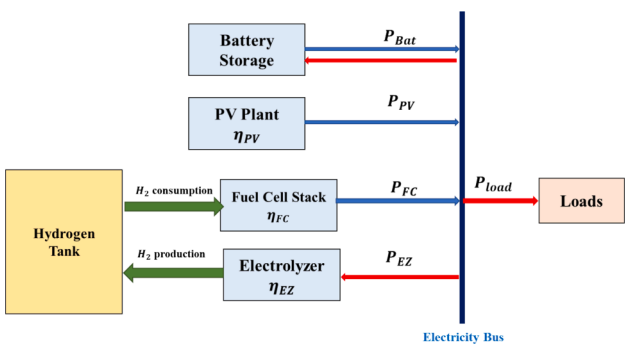


**Fig. 1.** Proposed isolated microgrid layout.

The BESS serves as a load, or an energy source based on generation from PV system and the loading of the microgrid. The electrolyzer consumes excess energy to produce hydrogen to store in the tank. This stored hydrogen can then be used to generate power for the microgrid via the fuel cell stack. Further details and mathematical representation of each subsystem is discussed in the next subsection.

### 2.1. PV plant

PV power generation is dependent mainly on solar irradiance. The model used for extracting the PV power generated $P_{PVi}$ as a function of solar irradiance $G_i$ at hour $i$ is [49]:

$$P_{PVi} = \frac{G_i}{1000} * \eta_{PV} * P_{PV,\,rated} \tag{1}$$

where $\eta_{PV}$ is the efficiency of the dc-dc converter and maximum power point tracking system, taken to be 95% [50]. $P_{PV,\,rated}$ is the rated power of PV plant, a decision variable found from the CC optimization. Since PV power generation acts as an input to the CC, it is formulated as a matrix with $N_s$ rows, representing the number of scenarios considered for the CC, and 24 columns representing the hourly steps of operation. $G_i$ as the solar irradiance at hour $i$ for a certain scenario. The solar irradiance data is obtained from historical datasets from [51]. The hourly solar irradiance data is fitted to a Gamma distribution. Then, the copula approach is used to model the uncertainties and dependencies of the random variables to account for solar irradiance uncertainties. This approach is useful in modelling the joint distributions, particularly when non-normal variables are involved [52].

### 2.2. Battery energy storage system

BESS is represented from stored energy amount point of view. For scenario $i$ and hour $j$, the amount of available energy stored after a certain hourly operation $E_{bat\,i,j}$ is:

$$E_{bat\,i,j} = E_{bat\,i,j-1} + P_{bat\,i,j} \cdot \Delta t \tag{2}$$

where $E_{bat\,i,j-1}$ is the available energy in the battery before that operation step, and $P_{bat\,i,j}$ is the amount of battery charging or discharging power from the battery. As in [38], the battery efficiency is neglected to reduce the model complexity and keep the chance constrained model as an MILP. $\Delta t$ is the length of time step, taken to be 1 *hour* in this work.

### 2.3. Hydrogen storage system

There are three components simultaneously involved in the operation of hydrogen system: the electrolyzer, fuel cell stack and the hydrogen tank. Electrolyzer is the equipment responsible for water chemical electrolysis to produce hydrogen and oxygen to be stored in sealed tanks under high pressures. Fuel cells are chemical devices that take pressurized hydrogen at the anode and oxygen at the cathode to produce electric power [53,54]. Hydrogen storage system mathematical model is based on [55]. The mathematical model is:

$$E_{H2tank\,i,j} = E_{H2tank\,i,j-1} + \left( \frac{P_{EZ\,i,j}}{\eta_{EZ}} - P_{FC\,i,j} \cdot \eta_{FC} \right) * \Delta t \tag{3}$$

where $E_{H2tank\,i,j}$ is the amount of hydrogen energy stored in the tank after the hour $(j-1)$ in scenario $i$. $E_{H2tank\,i,j-1}$ is the previously available energy in the tank. $P_{EZ\,i,j}$ is the decision taken for power to charge the hydrogen tank through the electrolyzer. $P_{FC\,i,j}$ is the power extracted from the fuel cell. $\eta_{EZ}$ and $\eta_{FC}$ are the operational efficiencies of the electrolyzer and the fuel cell stack, taken to be 80% and 60% respectively. These values are realistic with the imposed constraint on fuel stack power to be at least 60% of its nominal power [56,57]. Sizing the hydrogen tank, in *kWh* is achieved by utilizing hydrogen higher heating value, which represents the amount of energy released when 1 kg of hydrogen is combusted [58]. By multiplying the hydrogen size in kg by its higher heating value of 39.4 kWh/kg, conversion to kWh is achieved. The hydrogen tank size is expressed in kWh throughout the mathematical formulation to align with the energy-based optimization framework of the study.

### 2.4. Electrical load demand

Electrical load is the second uncertain parameter in the sizing process. In this work, both electrical load and solar irradiance data are obtained from a location in the UK. The dataset from Energy System Operator gives two load values for each hour [59]. For each hour of historical data, the two available data points are averaged for a single sample point per hour and converted to $kW$ levels instead of $GW$ original country-wide scale. The same process used for modelling the solar irradiance is used with load stochasticity modelling. A normal PDF is fit to the historical data of each hourly demand per season [24]. Copula is then used to model the uncertainties and dependencies in the load.

## 3. Methodology and problem formulation

### 3.1. Mathematical formulation of the CC optimization problem

The objective of this work is to minimize the total life cycle cost from all components of the microgrid. The Objective function is:

$$Min.Z = C_{capex} + C_{replacce} + C_{om} + C_{total,ls} + C_{total,og} + C_{penalty} \tag{4}$$

where $C_{capex}$ is the capital cost of system components $(USD)$, $C_{replacce}$ is the cost of replacement of system components, $(USD)$, $C_{om}$ is the cost of operation and maintenance $(USD)$, $C_{total,ls}$ is the total cost of shedding load $(USD)$, $C_{total,og}$ is the total cost of over generation $(USD)$, and $C_{penalty}$ is the penalty cost of violating the system constraints $(USD)$.

#### 3.1.1. Capital cost of system components

The major components of the microgrid system include the PV, the battery, the hydrogen tank, the electrolyzer and the fuel cell. The capital expenditure of these components is given by Eq. (5) and Eq. (6).

$$C_{capex} = C_{capex,PV} + C_{capex,bat} + C_{capex,FC} + C_{capex,EZ} + C_{capexH2tank} \tag{5}$$

$$C_{capex} = P_{PV,rated}.C_{PV} + E_{bat,cap}.C_{bat} + P_{FC,rated}.C_{FC} + P_{EZ,rated}.C_{EZ} + E_{H2tank}.C_{H2tank} \tag{6}$$

where $P_{PV,rated}$ is the PV Rated power decision variable, $kW$, $E_{bat,cap}$ is the battery energy capacity decision variable, $kWh$, $P_{FC,rated}$ is the fuel cell rated power decision variable, $kW$, $P_{EZ,rated}$ is the electrolyzer rated power decision variable, $kW$, $E_{H2tank}$ is the hydrogen tank size, $kWh$, and $C_k$ is the per unit cost of component $k$, values obtained from [55].

#### 3.1.2. Replacement cost of system components

To develop a more accurate model of the microgrid, the cost of replacement of the system components needs to be accounted for. This is because different components within the system have different lifetimes. Including the replacement cost therefore gives a true idea of the total system costs over the lifetime of the overall system. The replacement cost of the system components is given as:

$$C_{replacce} = n_{rep,PV}.C_{capex,PV} + n_{rep,bat}.C_{capex,bat} + n_{rep,FC}.C_{capex,FC} + n_{rep,EZ}.C_{capex,EZ} + n_{rep,H2tank}.C_{capexH2tank} \tag{7}$$

where $n_{rep,PV}$, $n_{rep,bat}$, $n_{rep,FC}$, $n_{rep,EZ}$, $n_{rep,H2tank}$ are the levelized parameters for the replacement cost of the PV, batteries, fuel cell, electrolyzer and the hydrogen tanks, respectively. These parameters relate the lifetime of the various components to the total system lifetime.

#### 3.1.3. Operational and maintenance cost

The day-to-day operation of the system would incur some costs. These include the costs incurred in cleaning PV modules, hydrogen tank and compressor maintenance, and safety and insurance costs. The operational and maintenance cost of the system is given by Eq. (8).

$$C_{om} = n_{om,sys}(C_{capex,PV} + C_{capex,bat} + C_{capex,FC} + C_{capex,EZ} + C_{capexH2tank}) \tag{8}$$

#### 3.1.4. Cost of over generation

The over generation of power is assigned a cost. This is because the generation of power involves the use of resources. If the total generation exceeds the load and storage capacity by a substantial amount, power quality and system security issues could arise. A margin of 10 % is allowed in this work. The cost of over generation is modeled as:

$$C_{total,og} = (\Delta E_{og,i,j} + \Delta E_{H2,i,j} + \Delta E_{bat,i,j}).\frac{365}{N_s.\delta}.n_{total,sys}.C_{og} \tag{9}$$

where $\Delta E_{og,i,j}$ is the positive difference in power generation and load demand at scenario $i$ and hour $j$, $\Delta E_{H2}$ is the positive difference in the final energy states and initial energy states of the hydrogen tank in $N_s$ scenarios, $\Delta E_{bat,i,j}$ is the positive difference in the final energy states and initial energy states of the battery in $N_s$ scenarios, $C_{og}$ is the cost of over generation per kWh, and $\delta$ is the success probability.

#### 3.1.5. Cost of load shedding

In situations where the generated power in the system is insufficient, it is required to shed some load to protect the system. This could be modelled as:

$$C_{total,ls} = (E_{g,i,j} - load_{i,j}).\frac{365}{N_s.\delta}.n_{total,sys}.C_{ls} \tag{10}$$

where $E_{g,i,j}$ is the energy generation in scenario $i$ and hour $j$ , and $load_{i,j}$ is the load demand in scenario $i$ and hour $j$. $C_{ls}$ is the cost of shedding load $(USD)$.

#### 3.1.6. Penalty cost

To achieve the required level of success for the constraints, a penalty cost is implemented for any violation of the constraints.

### 3.2. CC framework

CC optimization is a probabilistic technique suitable for problems involving stochastic variables. The main characteristic of a CC optimization approach is that instead of achieving the constraints, problem constraints are reformulated, such that the probability of achieving the constraints is constrained instead. For instance, instead of the general constraint $(F(x,y) \leq 0)$, CC optimization constraint becomes $(P(F(x,y) \leq 0) \geq \delta)$, such that $P$ is the probability function and $\delta$ is a parameter from 0 to 1, selected to ensure a certain minimum probability of achieving the required constraint. Formulating the CC problem and solving it analytically is very exhaustive and complicated to solve. Instead, simulation-based criteria is used [36]. In simulation-based CC, the number of scenarios $N_s$ is obtained for the variables involving uncertainties. Required success rate $\delta$ determines the number of cases needed to be feasible from $N_s$ scenarios. The algorithm ensures component selection is valid for the required portion of scenarios; $\delta^* N_s$. A MILP programming is then formulated with $N_s$ binary variables $u_i$, such that 1 represents the scenario where $i$ is successfully covered.

CC optimization constraints are transformed to a standard form. Each constraint is first transformed to the general form at step or hour $j$:

$$F\left(x_j, y_j\right) \leq 0 \tag{11}$$

Then, the constraint is reformulated for each scenario $i$, such that the binary $u_i$ variables determine achieving or fathoming this scenario:

$$F\left(x_{ij}, y_{ij}\right) \leq M \times (1 - u_i) \tag{12}$$

Such that $i \in \{1, 2, 3, \cdots\cdots.N_s\}$, $j \in \{1, 2, 3, \cdots\cdots.24\}$, $u_i \in \{0, 1\}$, and $M$ is a very large number.

### 3.2.1. Constraints formulation

This section describes the various constraints for the optimization of the developed problem. It is worth noting that these constraints take the form of Eq. (12) during implementation. The residual battery energy remaining during each hour and scenario is given by Eq. (13).

$$SOC_{min}{}^{*}E_{bat,cap} \leq E_{bat\,i,j} \leq SOC_{max}{}^{*}E_{bat,cap} \tag{13}$$

Here, $SOC_{min}$ and $SOC_{max}$ are the minimum and maximum percentages of battery energy capacity to remain always in batteries, and $E_{bat\,i,j}$ is the amount of energy in batteries after each hour $j$ in scenario $i$. By enforcing these bounds, the SOC is inherently represented in the model, as the battery energy level directly corresponds to the battery's SOC when normalized by the battery capacity to be sized. This ensures that the battery energy remains within specified operational limits, preventing overcharging or deep discharging, while allowing the optimization to focus on cost and performance without the added complexity of nonlinear SOC dynamics, maintaining the problem as MILP. A similar constraint is used for limiting the amount of energy stored in the hydrogen tank given by Eq. (14).

$$tank_{min}.E_{H2tank,} \leq E_{H2tank\,i,j} \leq tank_{max}.E_{H2tank} \tag{14}$$

where $tank_{min}$ and $tank_{max}$ are the minimum and maximum percentages of hydrogen tank energy capacity to remain always in the tank, and $E_{H2tank\,i,j}$ is the amount of energy in hydrogen tank after each hour $j$ in scenario $i$.

The power balance for the system is given by Eq. (15). This constraint ensures that the power generation is larger than load power.

$$P_{PV\,i,j} + P_{bat\,i,j} + P_{FC\,i,j} - P_{EZ\,i,j} - P_{load\,i,j} \geq 0 \tag{15}$$

Eq. (16) enables keeping the battery charging or discharging powers $P_{bat\,i,j}$ within specifications recommended for batteries to increase its efficiency and decrease its degradation [60,61].

$$-\frac{E_{bat,cap}}{bat_{Plim}} \leq P_{bat_{i,j}} \leq \frac{E_{bat,cap}}{bat_{Plim}} \tag{16}$$

The efficiency of fuel cell varies with its loading. To justify the use of constant fuel cell efficiency, its power $P_{FC\,i,j}$ is limited to be at least more than 60% of its rated power to-be-sized. This is represented as Eq. (17)

$$0.6P_{FC,\,rated} \leq P_{FC\,i,j} \leq P_{FC,\,rated} \tag{17}$$

In addition, it is required that the electrolyzer power, $P_{EZ\,i,j}$, used to generate compressed hydrogen, be always less than the rated power. This constraint is given as shown in:

$$0 \leq P_{EZ\,i,j} \leq P_{EZ,\,rated} \tag{18}$$

We force the final amounts of energies stored in batteries and hydrogen tanks to be at least the same as the initial amounts. This is translated to the constraints in Eq. (19) and Eq. (20).

$$E_{bat\,i,24} \geq SOC_0{}^{*}E_{bat,cap} \tag{19}$$

$$E_{H2tank\,i,24} \geq tank_0{}^{*}E_{H2tank,} \tag{20}$$

$SOC_0$ and $tank_0$ are the battery and hydrogen tank initial energy level percentages of their rated capacities to be decided.

Constraint (21) ensures that the target of the scenario-based CC is achieved. It is expected that a minimum number of success rate, where $u_i = 1$, be achieved based on the selected value of $\delta$. It should also be noted that Eqs. (2) and Eq. (3) are used within the optimization model as states constraints.

$$\frac{u_1 + u_2 + \cdots\cdots + u_{N_s}}{N_s} \geq \delta \tag{21}$$

## 4. Results and discussion

The problem is formulated as a MILP problem via a CC optimization approach. The formulated problem is solved in MATLAB using the YALMIP optimization toolbox with CPLEX and Gurobi solvers. These commercial solvers are widely recognized for their robustness, computational efficiency, and ability to handle large-scale MILP problems. These choices ensure effective solutions to complex optimization problems while maintaining computational feasibility. The CPLEX solver was selected for its strong reputation in efficiently solving large-scale optimization problems with high precision and reliability, making it a robust choice for complex formulations. On the other hand, Gurobi, with its versatility and advanced features, including its ability to handle challenging optimization problems while providing detailed diagnostics and flexibility in model tuning, make it suitable for achieving both robust and accurate solutions. The YALMIP optimization toolbox offers an excellent approach to solve MIP problems [62]. The specification if the device used for the simulation is Intel(R) Core (TM) i7-10510U CPU@1.80 GHz 2.3 GHz, NVIDIA GeForce MX130 GPU.

The parameters used in this work are presented in Table 2.

The optimization process aims to determine the optimal sizing of microgrid components by balancing system performance, reliability, and cost. It takes $N_s$ scenarios of hourly irradiance and load profiles as inputs. These scenarios represent the variability in renewable energy generation and demand over time, ensuring that the optimization accounts for diverse operational conditions. Using the parameters in Table 2, the formulated problem in Section 3 is solved to minimize the lifecycle cost shown in Eq. (4). The optimization generates outputs that include the optimal ratings of the battery, fuel cells, electrolyzer, and hydrogen tank. Theses outputs should maintain the formulated constraints. Out of the $N_s$ input scenarios, the ratings of the components must enable the system to operate successfully under at least $\delta \times N_s$ scenarios as stated in Eq. (21).

To implement the developed model, 100 scenarios are generated for the load and irradiance using copulas, which effectively capture the correlation between variables and enable realistic scenario generation. These load and irradiance scenarios are visualized in Fig. 2 and Fig. 3, respectively. Fig. 2(a) highlights specific cases, including those with maximum, minimum, 25$^{th}$, 50$^{th}$, and 75$^{th}$ percentile aggregated daily loads as annotated, along with 10 additional representative cases that provide a diverse view of potential variations. Fig. 2(b) provides an overview of the average hourly load across all scenarios, complemented by a box plot that captures the distribution of loads at each hour. The box plot shows the distribution of the 100 scenarios for each hour, with the bottom, middle and top of the box representing the 25$^{th}$, 50$^{th}$, and 75$^{th}$ percentiles of the data. The box plot reveals that the hourly load distributions exhibit significant variation, particularly during peak demand hours, indicating higher uncertainty. The whiskers extend to the range of most values, demonstrating the variability inherent in the stochastic nature of the load profiles. Similarly, Fig. 3(a) highlights specific irradiance scenarios of aggregated daily irradiance levels with

**Table 2**
Parameters used in system.

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| $C_{PV}$ | 857 USD/kW [63] | $tank_{min}$ | 0.0 |
| $C_{bat}$ | 212 USD/kWh [63] | $tank_{max}$ | 1.0 |
| $C_{FC}$ | 667 USD/kW | $tank_0$ | 0.8 |
| $C_{EZ}$ | 1450 USD/kW [63] | $bat_{Plim}$ | 3.0 |
| $C_{H2tank}$ | 6 $/kWh | $\eta_{FC}$ | 0.6 [63] |
| $SOC_{min}$ | 0.2 [64] | $\eta_{EZ}$ | 0.8 [15] |
| $SOC_{max}$ | 0.8 [64] | $N_s$ | 100 |
| $SOC_0$ | 0.3 | $\delta$ | 0.8 |
| $life_{PV}$ | 25 years [65] | $life_{bat}$ | 5 years [66] |
| $life_{FC}$ | 10 years [63] | $life_{EZ}$ | 10 years [63] |
| $life_{H2tank}$ | 20 years | $n_{total,sys}$ | 25 years |

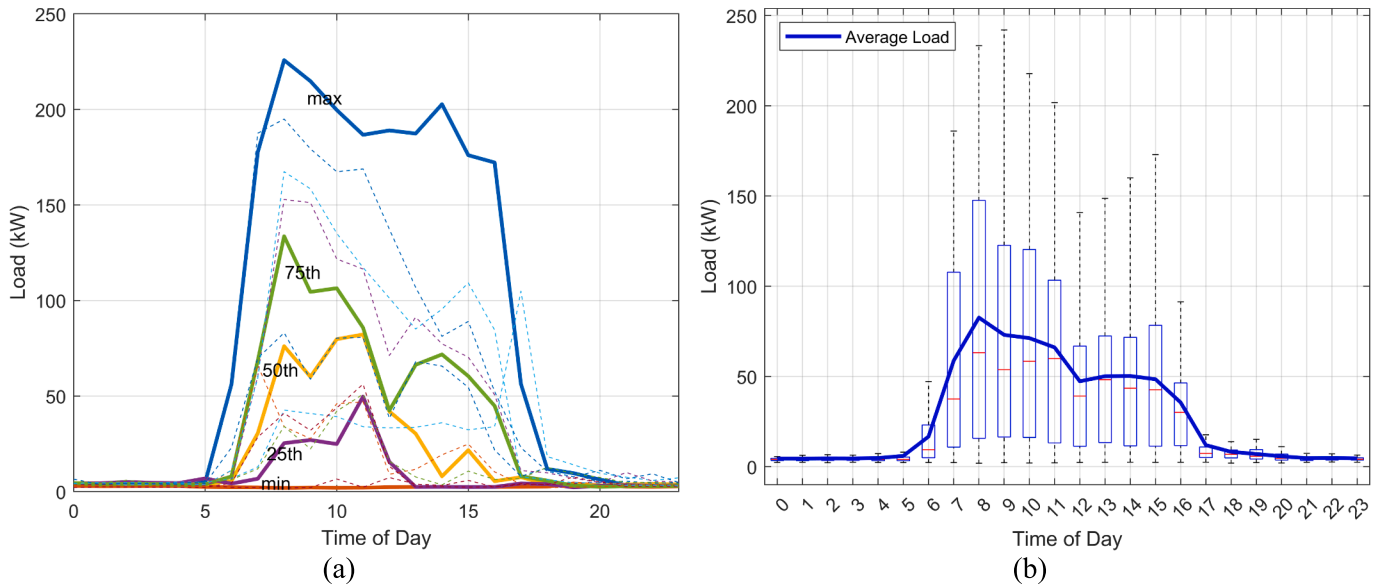


**Fig. 2.** Visualization of the 100 load scenarios generated: (a) Selected scenarios, including those with maximum, minimum, 25th, 50th, and 75th percentile daily loads, along with 10 additional representative cases. (b) Average hourly load and box plot of each hour.

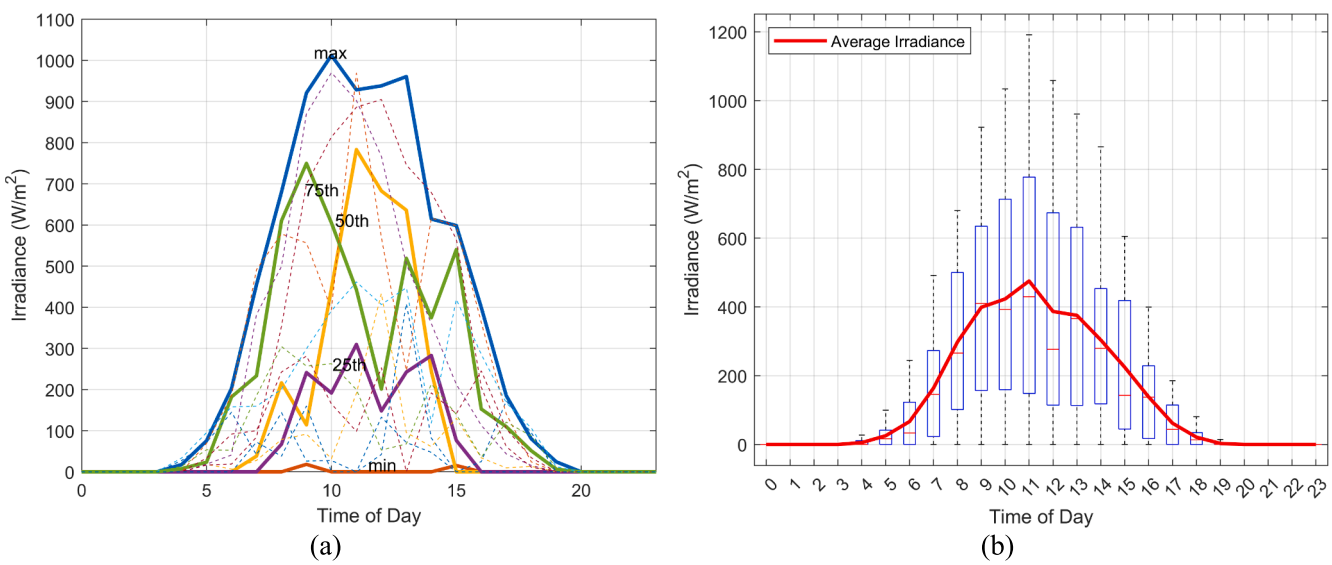


**Fig. 3.** Visualization of the 100 irradiance scenarios generated: (a) Selected scenarios, including those with maximum, minimum, 25th, 50th, and 75th percentile daily irradiance, along with 10 additional representative cases. (b) Average hourly irradiance and box plot of each hour.

10 additional representative cases. Fig. 3(b) presents the average hourly irradiance across all scenarios, accompanied by a box plot that illustrates the variability of irradiance values at each hour. The irradiance distributions reflect the typical diurnal pattern of solar energy, with peak irradiance occurring around midday. However, the box plot also highlights significant variability in irradiance during morning and late afternoon hours, where weather conditions such as cloud cover are likely to introduce uncertainty. The generated scenarios play a critical role in the chance-constrained optimization framework, as they capture a broad spectrum of possible conditions under which the microgrid must operate. The inclusion of scenarios with extreme conditions, such as maximum and minimum load or irradiance, ensures that the system design is robust and capable of handling diverse operational challenges. Moreover, the wide range of variability observed in the box plots underscores the importance of designing a system that balances reliability and cost-effectiveness. By considering this variability, the developed model can better accommodate uncertainties in load demand and solar energy availability.

Two solvers, CPLEX and Gurobi via YALMIP toolbox, are used to solve the problems and compared. With 100 scenarios, it is expected that the success rate $\delta$ be at least for 80 scenarios. Figs. 4-7 show the battery energy storage and hydrogen energy storage evolution within the day for each scenario of the successfully achieved scenarios for the two solvers. The results obtained using the CPLEX solver for the battery energy storage and hydrogen energy tank profiles are illustrated in Fig. 4 and Fig. 5, respectively. These figures provide insights into the system's ability to meet the operational constraints and handle variability across the successful scenarios. The battery energy profile demonstrates adherence to the defined operational constraints, with the energy levels consistently bounded by the maximum and minimum battery limits. The mean hourly profile shows a characteristic charging pattern during midday when solar irradiance is at its peak, followed by a gradual discharge during the evening and night to meet the load demand. The shaded region between the 25th and 75th percentiles represent the

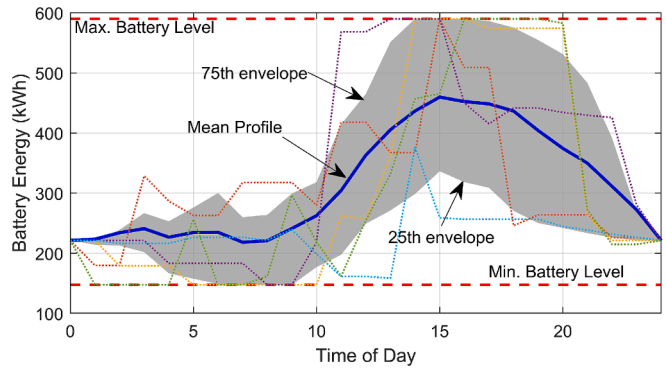


**Fig. 4.** Battery energy profile showing the mean hourly profile, 25th to 75th percentiles envelope and selected scenarios using CPLEX solver.

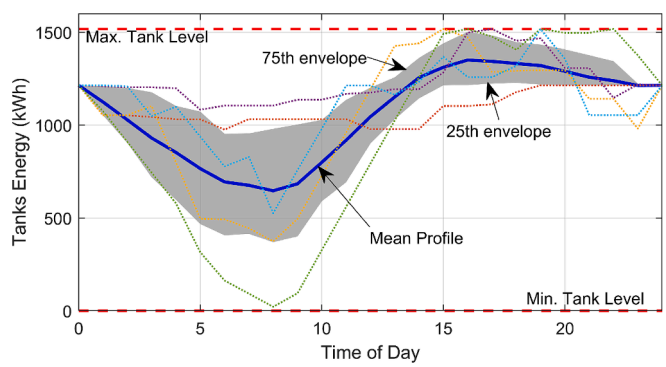


**Fig. 5.** Hydrogen energy tank profile showing the mean hourly profile, 25th to 75th percentiles envelope and selected scenarios using CPLEX solver.

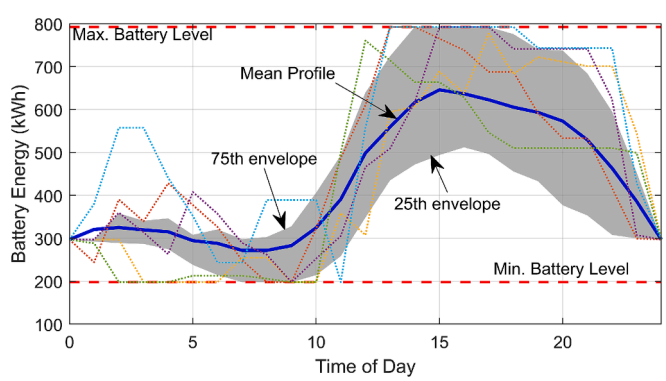


**Fig. 6.** Battery energy profile showing the mean hourly profile, 25th to 75th percentiles envelope and selected scenarios using Gurobi solver.

interquartile range of energy storage across the scenarios, capturing the variability in system operation due to stochastic inputs. This variability is more pronounced during the midday charging phase and the evening discharge period, reflecting the influence of fluctuating solar generation and varying load requirements. The presence of individual scenario trajectories outside the shaded region emphasizes the importance of including diverse scenarios in the optimization to ensure robustness under extreme conditions.

The hydrogen energy tank profile similarly using CPLEX satisfies the operational constraints, with energy levels confined within the maximum and minimum tank limits as seen in Fig. 5. The mean hourly profile reveals a gradual decline in hydrogen levels during the early hours of the day, corresponding to its utilization in meeting load demand when solar generation is insufficient. This is followed by a recovery

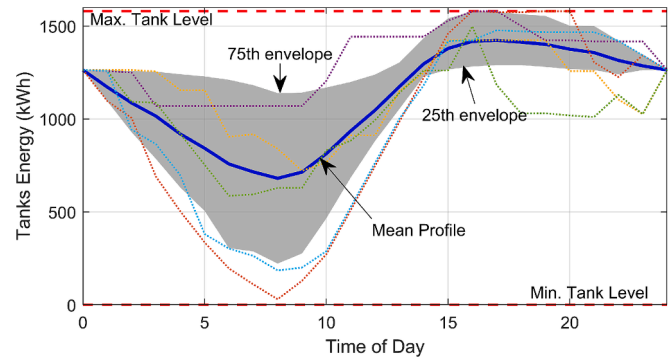


**Fig. 7.** Hydrogen energy tank profile showing the mean hourly profile, 25th to 75th percentiles envelope and selected scenarios using Gurobi solver.

phase during the late afternoon and evening as excess solar energy is converted and stored as hydrogen. The interquartile range, indicated by the shaded region, highlights significant variability during the recovery phase, driven by differences in solar availability and load profiles across scenarios. The broader range observed during this period suggests a higher sensitivity of hydrogen storage to input uncertainties compared to the battery. Despite this variability, the system successfully maintains hydrogen levels within the operational constraints, showcasing the efficacy of the CPLEX solver in ensuring feasibility under stochastic conditions.

The battery energy storage and hydrogen energy tank profiles generated using the Gurobi solver are presented in Fig. 6 and Fig. 7, respectively. These figures provide an overview of the system's operation under the influence of uncertainty, showcasing how the solver maintains feasibility while adhering to operational constraints. Like CPLEX, the battery energy profile solved by the Gurobi optimizer demonstrates strict compliance with the operational limits, with the energy levels consistently bounded between the minimum and maximum thresholds. The mean hourly profile reflects a typical pattern of midday charging, driven by surplus renewable energy, followed by a gradual discharge during peak load hours in the evening. The shaded envelope, representing the interquartile range, illustrates the variability across the scenarios. This variability is more prominent during midday and early evening, where the charging and discharging operations are heavily influenced by stochastic factors such as fluctuating solar generation and demand. Despite these variations, the profiles exhibit a controlled evolution of energy levels, reflecting the robustness of the optimization process in managing uncertainty.

For the hydrogen energy tank, the results from the Gurobi solver also adhere strictly to the operational constraints, with energy levels consistently within the minimum and maximum tank limits. The mean hourly profile reveals a steady reduction of stored hydrogen during the early hours of the day, followed by an increase phase during the afternoon and evening. This dynamic aligns with the system's need to balance short-term demand deficits with surplus energy conversion during periods of excess renewable generation. The variability captured within the interquartile range shows a significant spread during the recovery phase, similar to the results obtained with the CPLEX solver. This broader variability reflects the impact of scenario-specific differences in renewable generation and system demand. Nonetheless, the Gurobi solver ensures that the tank energy levels remain feasible and operationally sound, highlighting its reliability under uncertain conditions.

The overall behaviour of the system is consistent between the two solvers, with an overall success rate being 83 % and 84 % using the CPLEX and Gurobi solvers respectively, above the set threshold of 80 % set for the developed model. The battery serves as a dynamic short-term energy storage solution, effectively balancing fluctuations in demand and renewable energy supply throughout the day. Meanwhile, the hydrogen tank operates as a long-term energy buffer, handling gradual

energy deficits and surpluses over extended periods. Both solvers ensured that the storage units transitioned smoothly between charging and discharging phases, reflecting their ability to address the stochastic nature of the problem. One notable consistency between the solvers is their ability to capture variability across scenarios, as depicted by the interquartile range. Both CPLEX and Gurobi showed greater variability during midday and evening hours, which are periods of high uncertainty due to fluctuations in renewable energy generation and system demand. Despite this variability, the solvers successfully maintained feasibility, ensuring the system's reliable operation under all simulated conditions. Although both solvers performed well, some subtle differences in their results are worth noting. The profiles generated by the Gurobi solver showed slightly higher variability in both the battery and hydrogen tank operations during certain periods compared to CPLEX. This could indicate minor differences in how the solvers handle scenario-specific trade-offs, particularly in stochastic optimization environments. The results from both solvers suggest that integrating diverse energy storage technologies, like batteries and hydrogen tanks, can enhance system resilience by addressing both short-term and long-term uncertainties.

It is expected that the amount of generation balances the load demand in the system. This load consists of the actual system load and the demand from the energy storage components depending on their states. Fig. 8 and Fig. 9 illustrate the interaction between PV generation and the system's load demand load for sample scenarios using CPLEX and Gurobi solvers. In an ideal scenario, the renewable PV generation would match or exceed the load demand, which includes both the system's actual energy needs and the demands from the energy storage components. However, these figures reveal notable periods when PV generation is insufficient. In Fig. 8, under the CPLEX solver, insufficient generation is observed during the early morning hours between 0 and 7 h, around midday at approximately 12 h, and again in the evening between 15 and 23 h. Similarly, Fig. 9, representing the Gurobi solver sample scenario, shows several intervals during the day when PV generation fails to meet the load, highlighting the intermittency of renewable energy sources and their inability to ensure consistent supply. These shortage in generation emphasizes the importance of the batteries and hydrogen storages, which act as critical buffers to bridge the energy gap during these periods of inadequacy. However, there are multiple scenarios where the load had to be curtailed due to insufficient power supply from the PV and the storage components as shown in Figs. 10 and 11. Under the CPLEX solver, Fig. 10 demonstrates sporadic yet significant load curtailments, with a scenario reaching up to 136 kW of curtailed demand. In contrast, Fig. 11, representing the Gurobi solver, shows a similar pattern of load curtailment across scenarios but with

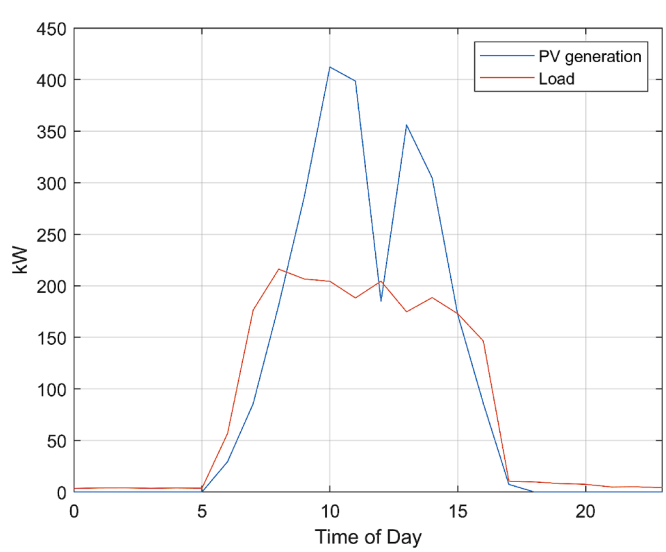


**Fig. 8.** Sample PV generation and load scenario from CPLEX.

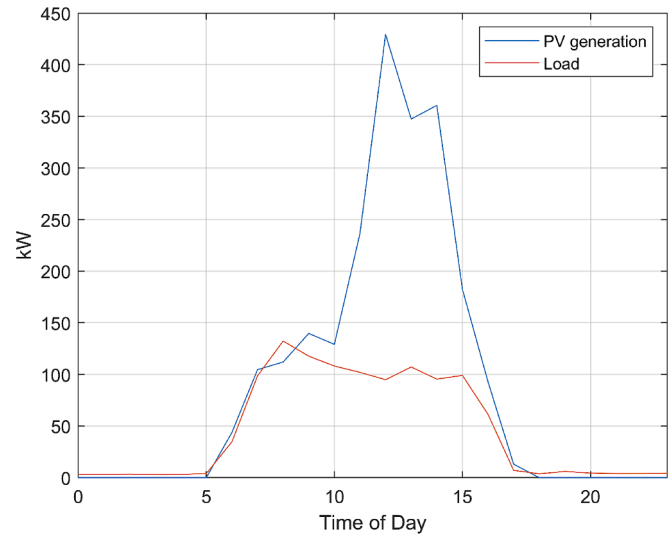


**Fig. 9.** Sample PV generation and load scenario from Gurobi.

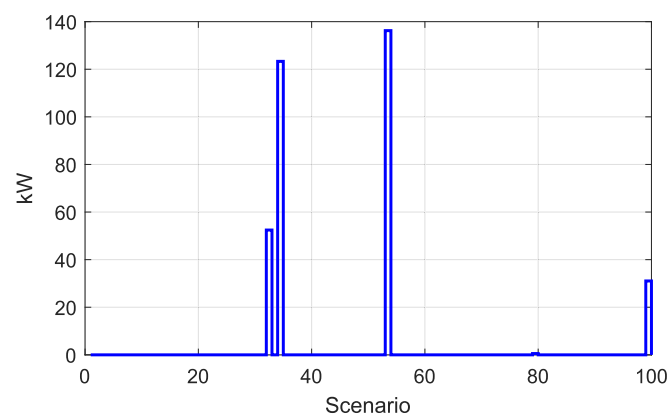


**Fig. 10.** Load shed using CPLEX across scenarios.

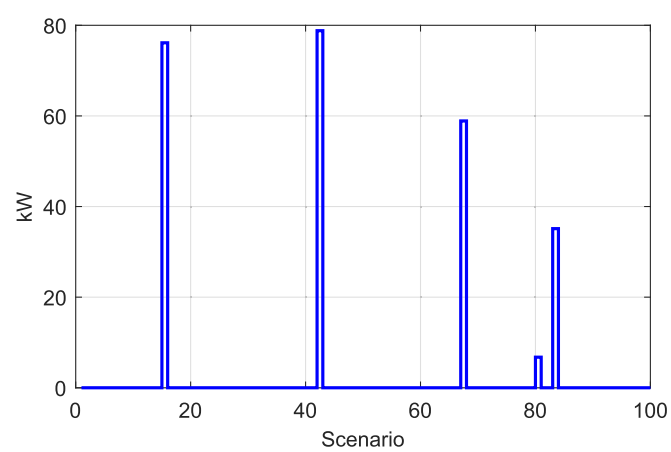


**Fig. 11.** Load shed using Gurobi across scenarios.

slightly lower maximum values, reaching up to 79 kW. This difference highlights variations in the solvers' optimization techniques and their approach to managing constraints under stochastic conditions. While storage helps mitigate the frequency and magnitude of load curtailment, its limited capacity and intermittency of renewables necessitate a margin of flexibility when operating isolated microgrids. This flexibility may include strategies such as controlled load shedding during critical

periods to maintain system stability. Even with robust optimization approaches, achieving 100 % continuity of supply cannot be guaranteed in microgrids relying primarily on intermittent renewable sources.

A summary of the results from the two optimization solvers is shown in Table 3. It presents the sizing of the components of the microgrid such that the system would match its load requirements at a minimum of 80 % success rate under the 100 case scenarios generated for the system. It is realized, from comparing the results, that both solvers perform to a similar level and produce results that complement each other. However, here are notable differences in computational efficiency, component sizing, and overall system cost. Gurobi solver utilized a higher computational time as compared to CPLEX, highlighting CPLEX's relative advantage in terms of computational efficiency. Additionally, the total cost of the system was marginally lower when optimized using CPLEX, amounting to approximately $1.221 million compared to $1.239 million for Gurobi. This cost difference, though minor, emphasizes the potential economic benefits of using CPLEX in similar optimization problems. In terms of component sizing, some differences are observed between the results of the two solvers. For instance, Gurobi allocated a larger battery capacity and hydrogen tank capacity compared to CPLEX's sizing. This suggests that Gurobi solution prioritized larger storage capacities, possibly as a strategy to mitigate uncertainties in renewable generation and load demand. On the other hand, CPLEX produced slightly smaller component sizes, resulting in a more cost-efficient design while still meeting the required performance thresholds. The rated capacities for other components also vary between the solvers. CPLEX solution had a larger PV system compared to Gurobi, which might reflect different trade-offs in balancing generation capacity, storage, and system costs.

## 5. Conclusion

In this work, a CC optimization formulation for a hybrid PV system, battery storage and hydrogen energy system are presented and implemented in this work. Although formulating problems as a CC problem could be challenging and exhaustive to solve, it provides a great opportunity for dealing with the uncertainties involved with power system variables, such as solar power generation and load. The formulation included constraints for limiting the rate of consumption of both energy storage to reduce battery degrading and increase both systems efficiencies. Then, a mixed integer linear programming optimization was formulated to implement the CC problem. Two solvers, CPLEX and Gurobi were utilized to solve the developed problem. The objective of the optimization is to size the components of the microgrid such that the load demands of the system are met, and the total life cycle cost of the system is minimum. The results showed the capacities of the various components needed to meet the load requirements of a microgrid with a PV source, hydrogen storage system and a battery energy storage system. The total life cycle cost of the system for a period of 25 years is found to be US$ 1.221 $m$.

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude in order to ensure the writing is grammatically correct and avoid word repetition. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

## CRediT authorship contribution statement

**Ahmed R. El Shamy:** Conceptualization, Methodology / Study design, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization, Project administration. **Prince Aduama:** Writing – review & editing, Validation, Software, Project administration, Methodology, Investigation, Formal analysis. **Ameena S. Al-Sumaiti:** Methodology, Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization, Formal analysis, Data curation.

**Table 3**
Summary of results.

| Optimization Variable | CPLEX | GUROBI |
|---|---|---|
| $P_{PV,rated}(kW)$ | 597.1162 | 539.7330 |
| $E_{bat,cap}(kWh)$ | 737.3857 | 990.6818 |
| $P_{FC,rated}(kW)$ | 182.0872 | 189.5437 |
| $P_{EZ,rated}(kW)$ | 291.0537 | 296.8561 |
| $E_{H2tank}(kWh)$ | 1517.4000 | 1579.5000 |
| $E_{H2tank}(kg)$ | 38.5027 | 40.0794 |
| $Total\ Cost\ (USD)$ | $1.2210*10^6$ | $1.2390*10^6$ |
| Success rate ($\delta$) | 83 % | 84 % |
| Simulation time (s) | 1881.8500 | 2552.0870 |

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

This work is supported in part by ASPIRE under the ASPIRE Virtual Research Institute (VRI) Program, Award Number VRI20-07, and the ENGR 610 course at Khalifa University.

## Data availability

Used data are open source and referenced in text.

## References

[1] Akter H, Howlader HOR, Mamud F, Saber AY, Yona A, Senjyu T. Optimal sizing and performance analysis of hybrid microgrid for remote island of developing country: effect of sustainable parameters, benefits and installation barriers. *Franklin Open* 2024:100074.
[2] Al-Sumaiti AS, Ahmed MH, Rivera S, El Moursi MS, Salama MM, Alsumaiti T. Stochastic PV model for power system planning applications. *IET Renew Power Gener* 2019;13(16):3168–79.
[3] Lasseter RH, Paigi P. Microgrid: a conceptual solution. In: 2004 IEEE 35th annual power electronics specialists conference (IEEE Cat. No. 04CH37551), 2004, vol. 6, pp. 4285-4290: IEEE.
[4] Muhtadi A, Pandit D, Nguyen N, Mitra J. Distributed energy resources based microgrid: review of architecture, control, and reliability. *IEEE Trans Ind Appl* 2021;57(3):2223–35.
[5] Elsir M, Al-Sumaiti AS, El Moursi MS, Al-Awami AT. Coordinating the day-ahead operation scheduling for demand response and water desalination plants in smart grid. *Appl Energy* 2023;335:120770.
[6] Dong X-J, He G-X, Zhou Z-W, Shen J-N, He Y-J. Simultaneous design and scheduling optimization of the photovoltaic-wind-hydropower-hydrogen hybrid system. *Energ Conver Manage* 2024;314:118638.
[7] Faisal M, Hannan MA, Ker PJ, Hussain A, Mansor MB, Blaabjerg F. Review of energy storage system technologies in microgrid applications: issues and challenges. *IEEE Access* 2018;6:35143–64.
[8] Georgious R, Refaat R, Garcia J, Daoud AA. Review on energy storage systems in microgrids. *Electronics* 2021;10(17):2134.
[9] El-Hamalawy AF, Farag HE, Asif A. Optimal design of grid-connected green hydrogen plants considering electrolysis internal parameters and battery energy storage systems. *Energ Conver Manage* 2024;302:118127.
[10] Irena. Reaching zero with renewables: Eliminating CO2 emissions from industry and transport in line with the 1.5°C climate goal. Int Renew Energy Agency 2020.
[11] Zeng R, et al. Design and optimization of solar energy system with hydrogen energy storage and alkaline fuel cell. *Energ Conver Manage* 2023;295:117628.
[12] Polleux L, Guerassimoff G, Marmorat J-P, Sandoval-Moreno J, Schuhler T. An overview of the challenges of solar power integration in isolated industrial microgrids with reliability constraints. *Renewable Sustainable Energy Reviews* 2022;155:111955.
[13] Mathew M, Hossain MS, Saha S, Mondal S, Haque ME. Sizing approaches for solar photovoltaic-based microgrids: a comprehensive review. *IET Energy Syst Integr* 2022;4(1):1–27.

[14] Mohandes B, Acharya S, Moursi MSE, Al-Sumaiti AS, Doukas H, Sgouridis S. Optimal design of an islanded microgrid with load shifting mechanism between electrical and thermal energy storage systems. *IEEE Trans Power Syst* 2020;35(4):2642–57.

[15] Medghalchi Z, Taylan O. A novel hybrid optimization framework for sizing renewable energy systems integrated with energy storage systems with solar photovoltaics, wind, battery and electrolyzer-fuel cell. *Energ Conver Manage* 2023;294:117594.

[16] Colarossi D, Principi P. Optimal sizing of a photovoltaic/energy storage/cold ironing system: life cycle cost approach and environmental analysis. *Energ Conver Manage* 2023;291:117255.

[17] Mahian O, Javidmehr M, Kasaeian A, Mohasseb S, Panahi M. Optimal sizing and performance assessment of a hybrid combined heat and power system with energy storage for residential buildings. *Energ Conver Manage* 2020;211:112751.

[18] Jia K, Liu C, Li S, Jiang D. Modeling and optimization of a hybrid renewable energy system integrated with gas turbine and energy storage. *Energ Conver Manage* 2023;279:116763.

[19] Nurunnabi M, Roy NK, Hossain E, Pota HR. Size optimization and sensitivity analysis of hybrid wind/PV micro-grids-a case study for Bangladesh. *IEEE Access* 2019;7:150120–40.

[20] Diab AAZ, Sultan HM, Mohamed IS, Kuznetsov ON, Do TDJIA. Application of different optimization algorithms for optimal sizing of PV/wind/diesel/battery storage stand-alone hybrid microgrid 2019;7:119223–119245.

[21] Marqusee J, Becker W, Ericson S. Resilience and economics of microgrids with PV, battery storage, and networked diesel generators. *Adv Appl Energy* 2021;3:100049.

[22] Salman UT, Al-Ismail FS, Khalid M. Optimal sizing of battery energy storage for grid-connected and isolated wind-penetrated microgrid. *IEEE Access* 2020;8:91129–38.

[23] Wu X, Zhao W, Wang X, Li H. An MILP-based planning model of a photovoltaic/diesel/battery stand-alone microgrid considering the reliability. *IEEE Trans Smart Grid* 2021;12(5):3809–18.

[24] Ramadan A, Ebeed M, Kamel S, Abdelaziz AY, Haes Alhelou H. Scenario-based stochastic framework for optimal planning of distribution systems including renewable-based DG units. *Sustainability* 2021;13(6):3566.

[25] Ban M, Guo D, Yu J, Shahidehpour M. Optimal sizing of PV and battery-based energy storage in an off-grid nanogrid supplying batteries to a battery swapping station. J Mod Power Syst Clean Energy 2019;7(2):309–20.

[26] Aghamohamadi M, Mahmoudi A, Haque MH. Two-stage robust sizing and operation co-optimization for residential PV–battery systems considering the uncertainty of PV generation and load. IEEE Trans Ind Inf 2020;17(2):1005–17.

[27] Xie R, Wei W, Shahidehpour M, Wu Q, Mei S. Sizing renewable generation and energy storage in stand-alone microgrids considering distributionally robust shortfall risk. *IEEE Trans Power Syst* 2022.

[28] Furlan G, You F. Robust design of hybrid solar power systems: sustainable integration of concentrated solar power and photovoltaic technologies. *Adv Appl Energy* 2024;13:100164.

[29] Zhao N, You F. New York State's 100% renewable electricity transition planning under uncertainty using a data-driven multistage adaptive robust optimization approach with machine-learning. *Adv Appl Energy* 2021;2:100019.

[30] Oyewole OL, Nwulu NI, Okampo EJ. Optimal design of hydrogen-based storage with a hybrid renewable energy system considering economic and environmental uncertainties. *Energ Conver Manage* 2024;300:117991.

[31] Lai CS, McCulloch MD. Sizing of stand-alone solar PV and storage system with anaerobic digestion biogas power plants. *IEEE Trans Ind Electron* 2016;64(3):2112–21.

[32] Hakimi SM, Hasankhani A, Shafie-khah M, Lotfi M, Catalao JP. Optimal sizing of renewable energy systems in a Microgrid considering electricity market interaction and reliability analysis. *Electr Pow Syst Res* 2022;203:107678.

[33] Polimeni S, Meraldi L, Moretti L, Leva S, Manzolini G. Development and experimental validation of hierarchical energy management system based on stochastic model predictive control for Off-grid Microgrids. *Adv Appl Energy* 2021;2:100028.

[34] Ju L, et al. Two-stage optimal dispatching model and benefit allocation strategy for hydrogen energy storage system-carbon capture and utilization system-based micro-energy grid. *Energ Conver Manage* 2024;313:118618.

[35] Boyd S, Boyd SP, Vandenberghe L. Convex optimization. Cambridge university press; 2004.

[36] Calafiore GC, Campi MC. The scenario approach to robust control design. *IEEE Trans Autom Control* 2006;51(5):742–53.

[37] Margellos K, Goulart P, Lygeros JJIToAC. On the road between robust optimization and the scenario approach for chance constrained optimization problems 59(8) (2014) 2258-2263.

[38] Geng S, Vrakopoulou M, Hiskens IA. Chance-constrained optimal capacity design for a renewable-only islanded microgrid. *Electr Pow Syst Res* 2020;189:106564.

[39] Dashti H, Cheng J, Krokhmal P. Chance-constrained optimization-based solar microgrid design and dispatch for radial distribution networks. *Energy Syst* 2022;13(4):959–81.

[40] Liu Z, Wen F, Ledwich G. Optimal siting and sizing of distributed generators in distribution systems considering uncertainties. *IEEE Trans Power Delivery* 2011;26(4):2541–51.

[41] Zare A, Chung C, Zhan J, Faried SO. A distributionally robust chance-constrained MILP model for multistage distribution system planning with uncertain renewables and loads. *IEEE Trans Power Syst* 2018;33(5):5248–62.

[42] Singh S, Chauhan P, Singh N. Capacity optimization of grid connected solar/fuel cell energy system using hybrid ABC-PSO algorithm. *Int J Hydrogen Energy* 2020;45(16):10070–88.

[43] García-Triviño P, Llorens-Iborra F, García-Vázquez CA, Gil-Mena AJ, Fernández-Ramírez LM, Jurado F. Long-term optimization based on PSO of a grid-connected renewable energy/battery/hydrogen hybrid system. *Int J Hydrogen Energy* 2014;39(21):10805–16.

[44] Yahya W, et al. Optimization of a hybrid renewable energy system consisting of a of PV/wind turbine/battery/fuel cell integration and component design. *Int J Hydrogen Energy* 2024;94:1406–18.

[45] Lv X, Li X, Xu C. A robust optimization model for capacity configuration of PV/battery/hydrogen system considering multiple uncertainties. *Int J Hydrogen Energy* 2023;48(21):7533–48.

[46] Coppitters D, De Paepe W, Contino F. Robust design optimization and stochastic performance analysis of a grid-connected photovoltaic system with battery storage and hydrogen storage. *Energy* 2020;213:118798.

[47] Mahmoud FS, Abdelhamid AM, Al Sumaiti A, El-Sayed A-H-M, Diab AAZ. Sizing and design of a PV-wind-fuel cell storage system integrated into a grid considering the uncertainty of load demand using the marine predators algorithm. *Mathematics* 2022;10(19):3708.

[48] S. Nojavan, M. Majidi, K. J. E. C. Zare, and Management, "Performance improvement of a battery/PV/fuel cell/grid hybrid energy system considering load uncertainty modeling using IGDT," *Energy Conversion and Management,* vol. 147, pp. 29-39, 2017.

[49] Rezaei E, Dagdougui H. Optimal real-time energy management in apartment building integrating microgrid with multizone HVAC control. *IEEE Trans Ind Inf* 2020;16(11):6848–56.

[50] Moosavi SKR, Mansoor M, Zafar MH, Khan NM, Mirza AF, Akhtar N. Highly efficient maximum power point tracking control technique for PV system under dynamic operating conditions. *Energy Rep* 2022;8:13529–43.

[51] Stackhouse P. NASA POWER | Data Access Viewer; 2022. Available: https://power.larc.nasa.gov/data-access-viewer/.

[52] Trivedi PK, Zimmer DMJF, Econometrics Ti. Copula modeling: an introduction for practitioners 2007;1(1):1–111.

[53] Fathy A. A reliable methodology based on mine blast optimization algorithm for optimal sizing of hybrid PV-wind-FC system for remote area in Egypt. *Renew Energy* 2016;95:367–80.

[54] Larminie J, Dicks A, McDonald MS. Fuel cell systems explained. J. Wiley Chichester, UK; 2003.

[55] Khiareddine A, Salah CB, Rekioua D, Mimouni MF. Sizing methodology for hybrid photovoltaic/wind/hydrogen/battery integrated to energy management strategy for pumping system. *Energy* 2018;153:743–62.

[56] Sharer P, Rousseau A. Benefits of fuel cell range extender for medium-duty vehicle applications. World Electric Veh J 2013;6(2):452–63.

[57] E. E. a. R. E. D. US Department of Energy, "Comparison of Fuel Cell Technologies: Fact Sheet," Available: https://www.energy.gov/eere/fuelcells/articles/comparison-fuel-cell-technologies-fact-sheet.

[58] ENAPTER. (01/12/2024). What is the energy content of hydrogen? Available: https://www.enapter.com/newsroom/kb_post/what-is-the-energy-content-of-hydrogen#:~:text=January%2011%2C%202022-,The%20energy%20content%20of%20hydrogen%20is%20described%20by%20its%20(lower,kg%20or%203.28%20kWh%2FNm%C2%B3.

[59] ESO. Electricity consumption UK 2009-2022; 2022.

[60] Saxena S, Xing Y, Kwon D, Pecht M. Accelerated degradation model for C-rate loading of lithium-ion batteries. J Int J Electr Power Energy Syst 2019;107:438–45.

[61] TESLA. Powerwall 2 AC Datasheet," ed: TESLA; 2019.

[62] Lofberg J. YALMIP: A toolbox for modeling and optimization in MATLAB. In: 2004 IEEE international conference on robotics and automation (IEEE Cat. No. 04CH37508). IEEE; 2004. p. 284-289.

[63] Li L, Sun Y, Han Y, Chen W. Seasonal hydrogen energy storage sizing: two-stage economic-safety optimization for integrated energy systems in northwest China. IScience 2024;27(9).

[64] Jang YJ, Jeong S, Lee MS. Initial energy logistics cost analysis for stationary, quasi-dynamic, and dynamic wireless charging public transportation systems. *Energies* 2016;9(7):483.

[65] Herceg S, Fischer M, Weiß K-A, Schebek L. Life cycle assessment of PV module repowering. *Energ Strat Rev* 2022;43:100928.

[66] Hoppmann J, Volland J, Schmidt TS, Hoffmann VH. The economic viability of battery storage for residential solar photovoltaic systems – a review and a simulation model. Renew Sustain Energy Rev 2014;39:1101–18.